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Modeling Pointing Tasks in Human-Blimp Interactions

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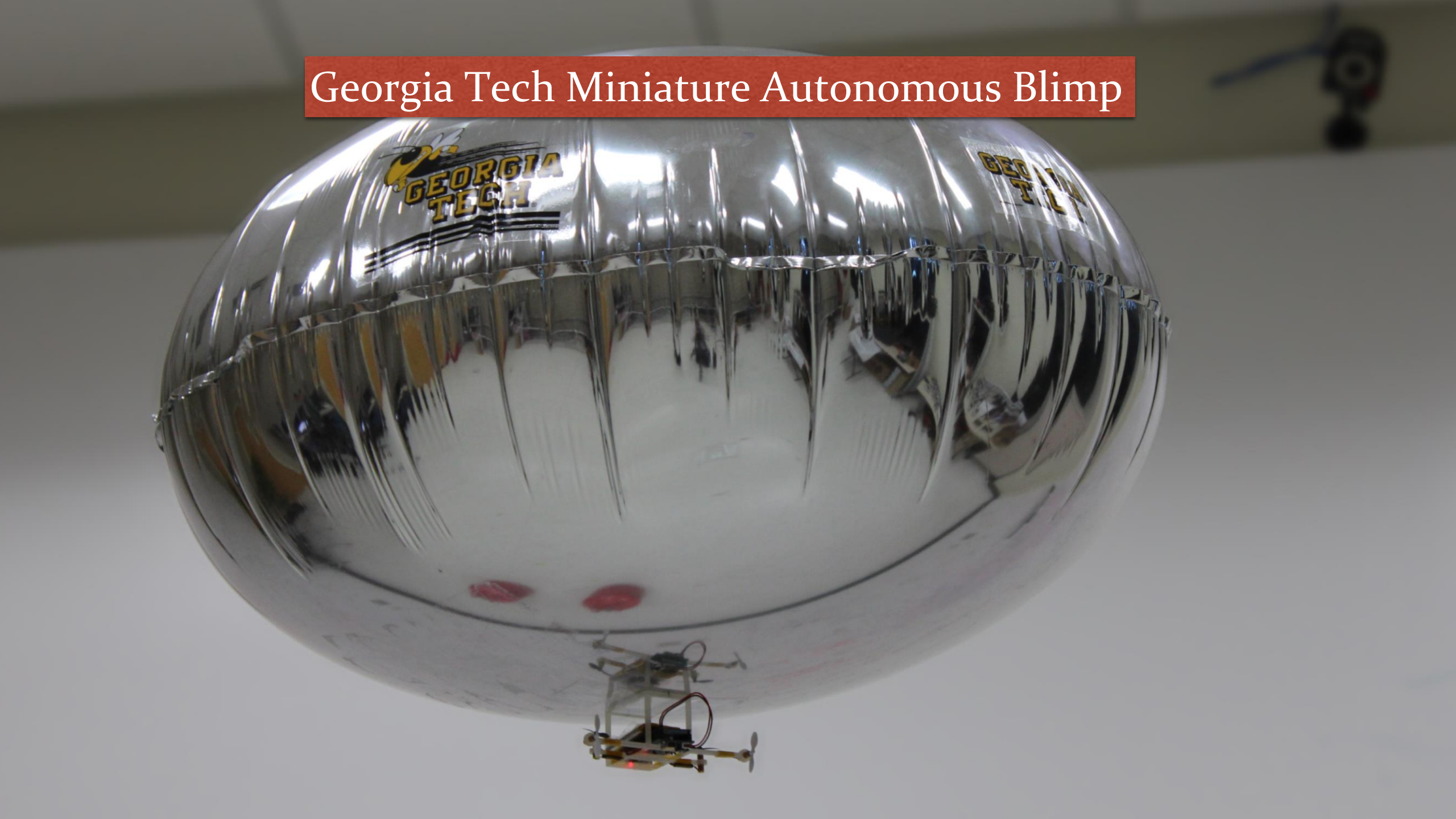
Georgia Tech Miniature Autonomous Blimp



GT-MAB: UAV platform for indoor human-occupied environment

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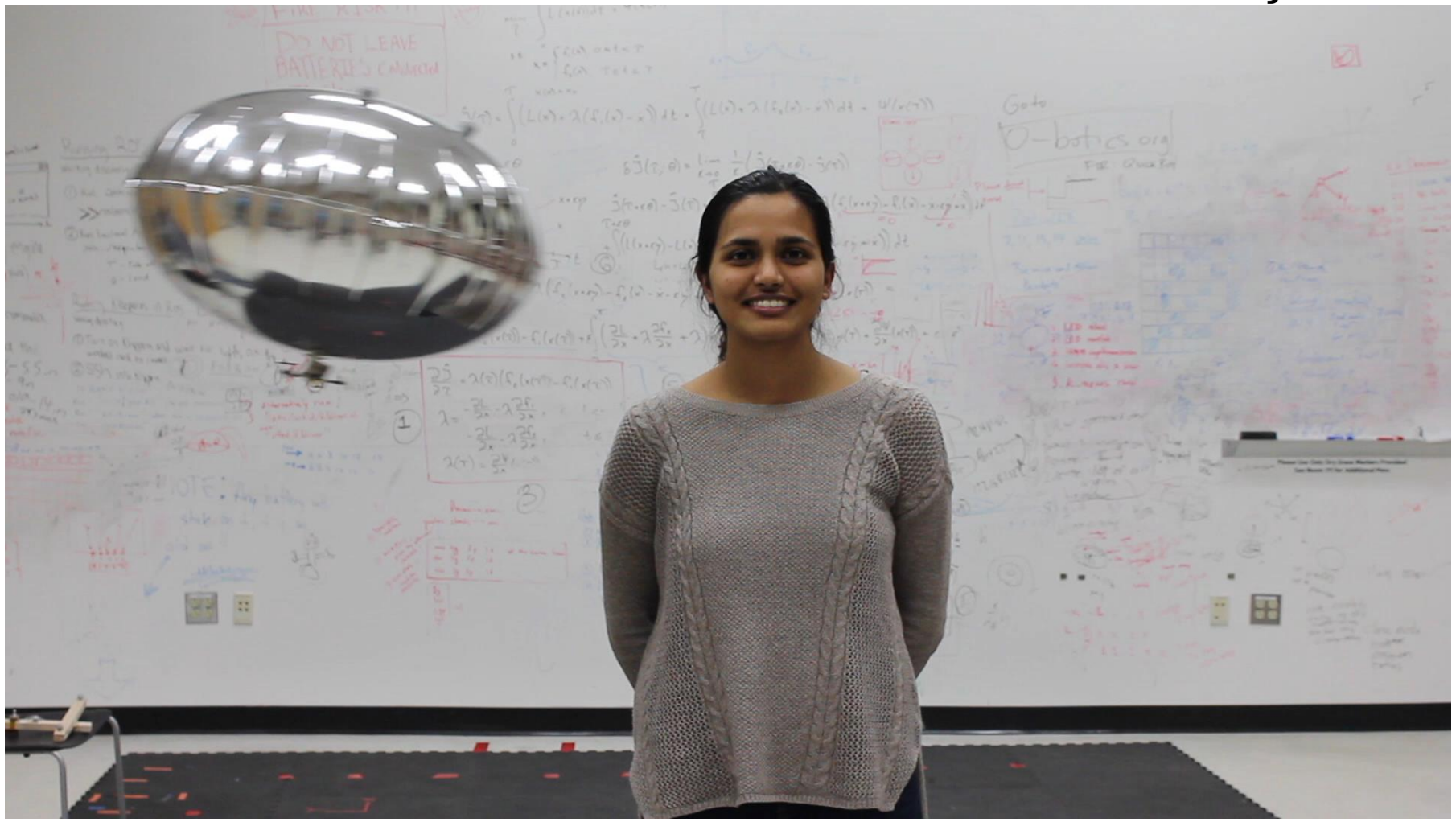

No safety threat to human

Extended endurance

HRI using hand movement

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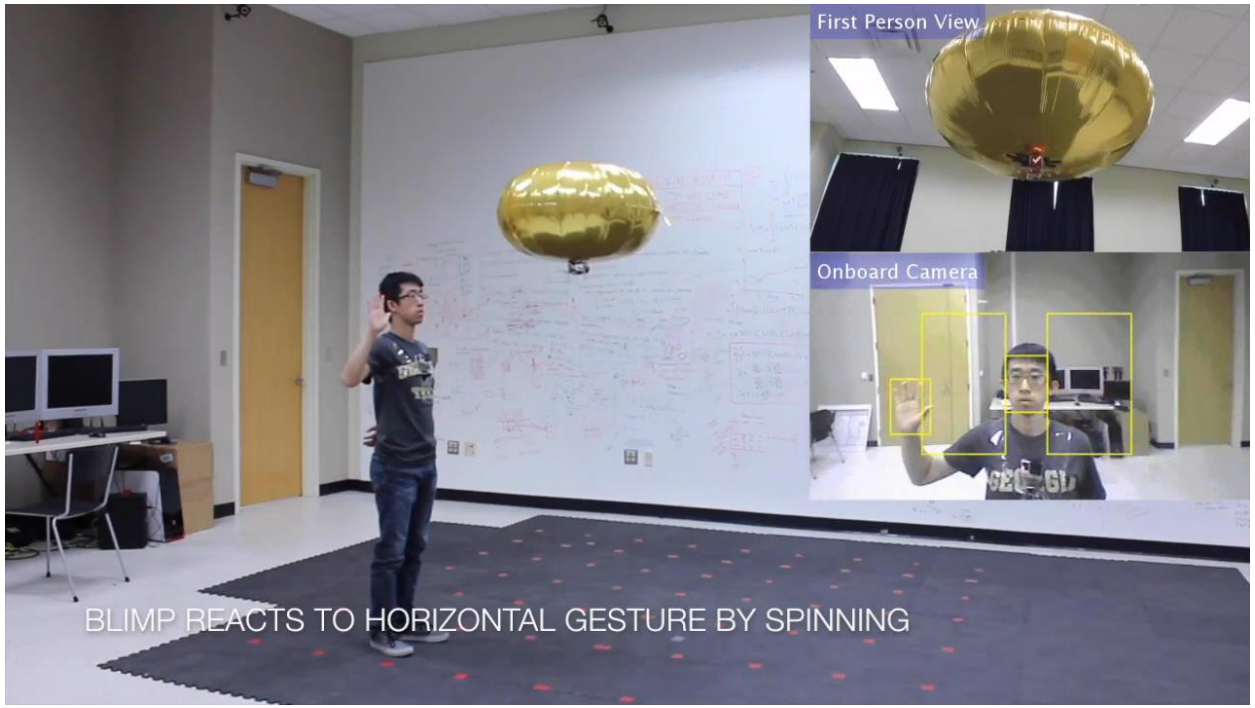
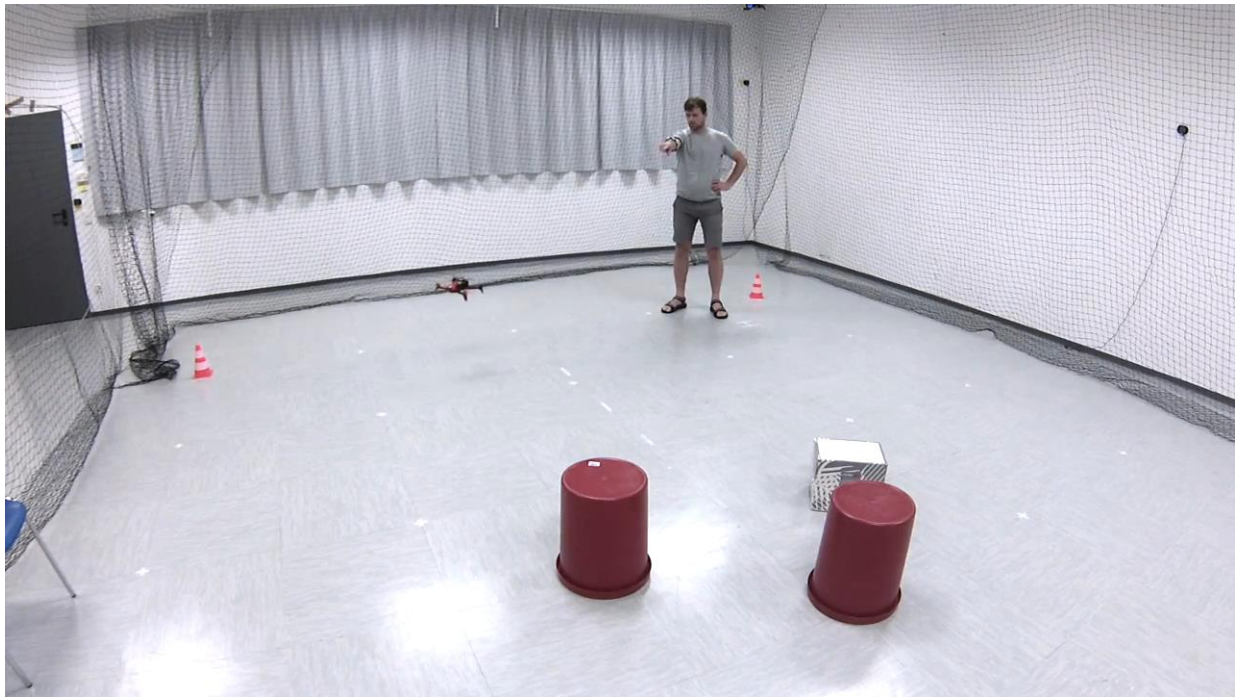


Hand gestures [1]

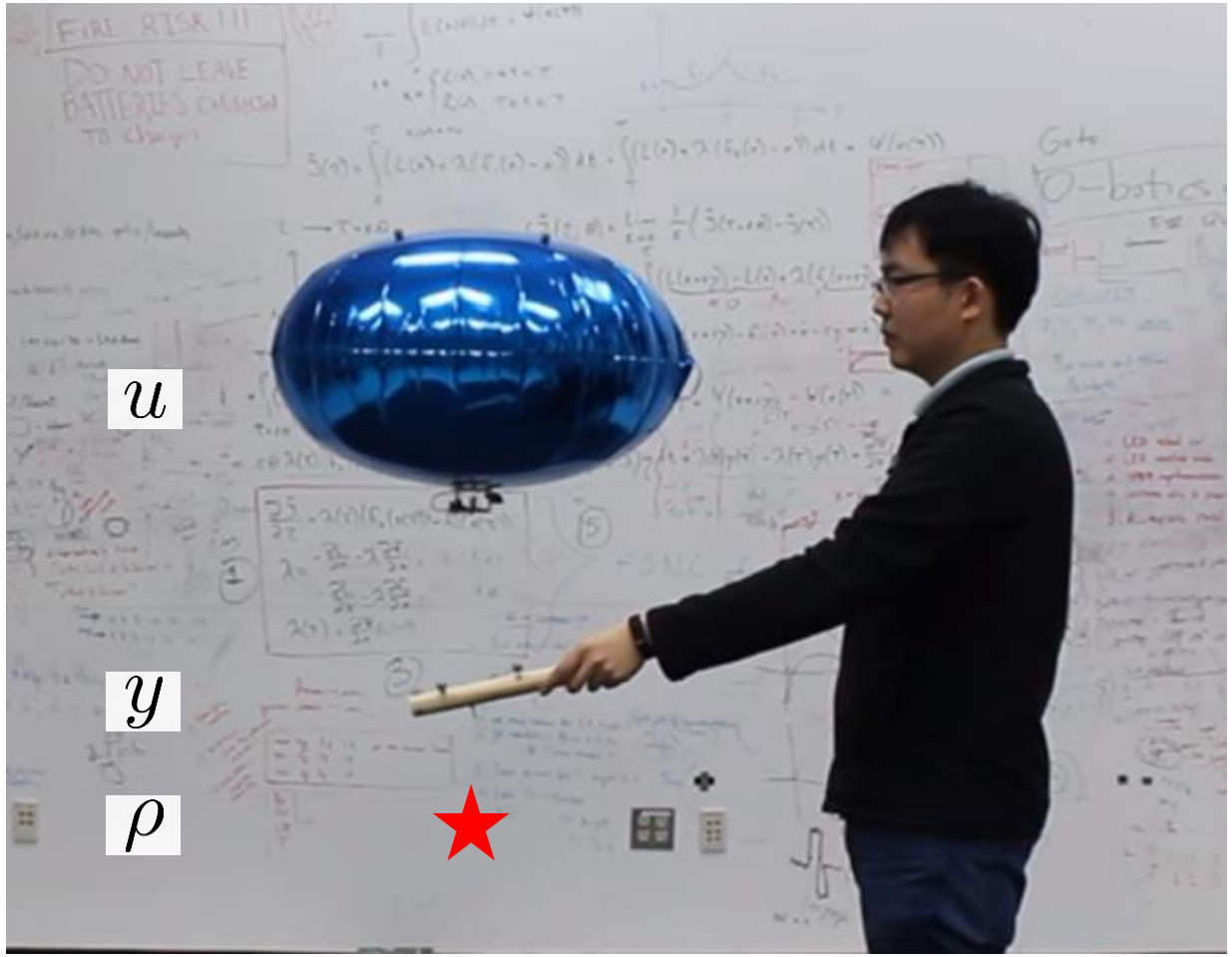


Pointing motion [2]

[1] N. Yao, et al. "Autonomous flying blimp interaction with human in an indoor space." Frontiers of Information Technology & Electronic Engineering 20.1 (2019): 45-59.

[2] B. Gromov, L. Gambardella, and A. Giusti, “Video: Landing a Drone with Pointing Gestures,” in *HRI ’18 Companion: 2018 ACM/IEEE International Conference on Human-Robot Interaction Companion, March 5–8, 2018, Chicago, IL, USA*, 2018.

Objective: modeling pointing motion in Human-Blimp interaction



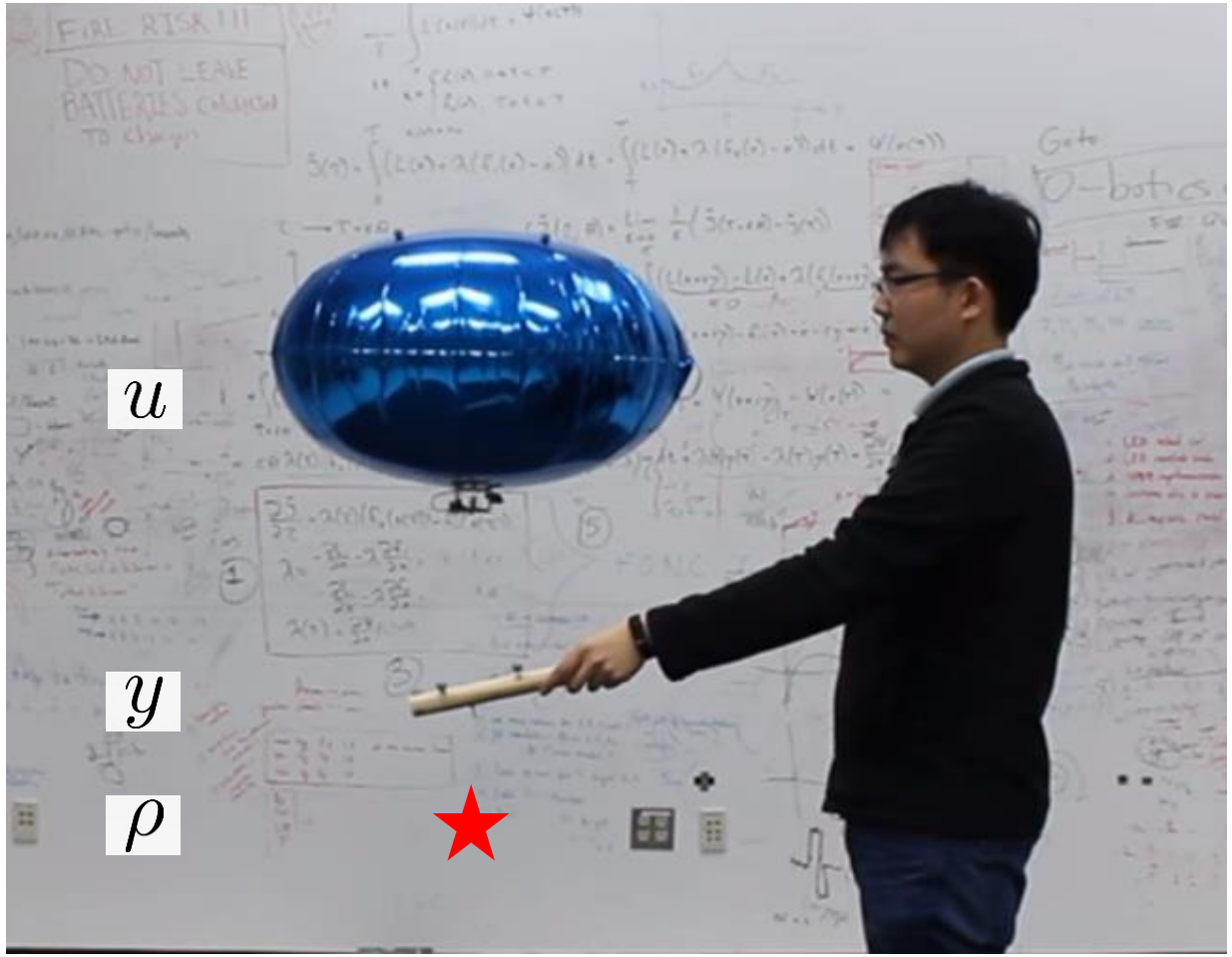
$\rho \in \mathbb{R}$: Target position

$u(t) \in \mathbb{R}$: Perceived blimp position

$y(t) \in \mathbb{R}$: Wand position

- Model dynamics of human's pointing motion when interacting with a flying object in 3D space
- Identify the intention of human subject (desired target position)

Objective: modeling pointing motion in Human-Blimp interaction



$\rho \in \mathbb{R}$: Target position

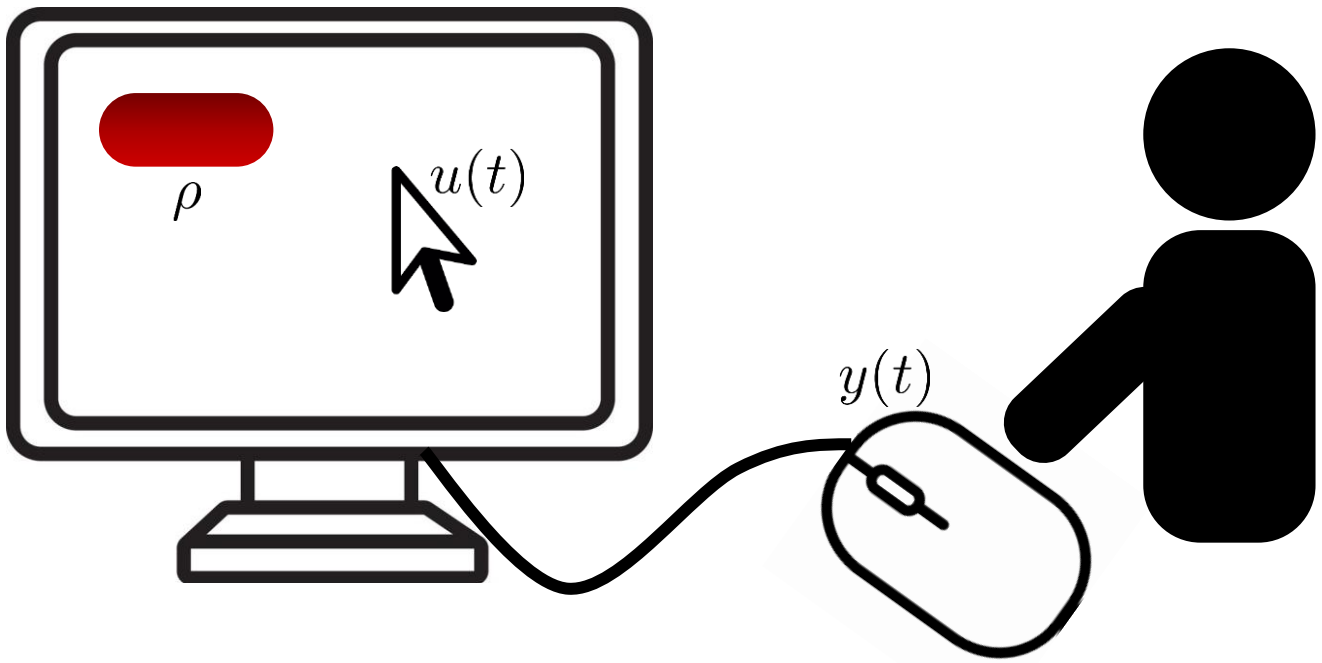
$u(t) \in \mathbb{R}$: Perceived blimp position

$y(t) \in \mathbb{R}$: Wand position

Challenges:

- No existing model describing the dynamics of human's pointing motion when interacting with a flying object in 3D space.
- No existing model describing human intention for human interacting with aerial robot.

VITE (Vector Integration to Endpoint) model

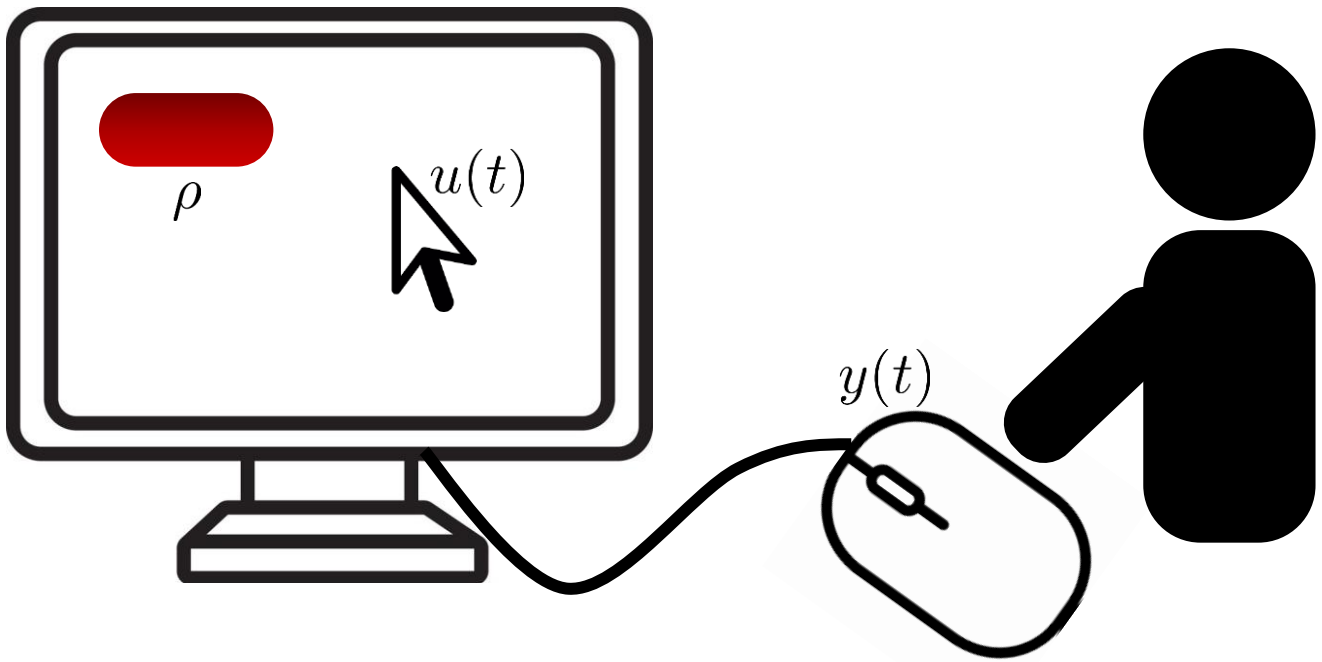


u: Perceived pointer position on screen

y: Pointer device position

ρ: Desired target position on screen

VITE (Vector Integration to Endpoint) model



If we consider the 1D pointing motion:

$$\begin{cases} \dot{\eta}(t) = \gamma(-\eta(t) + \rho - u(t)) \\ \dot{y}(t) = g[\eta(t)]_d^+ \end{cases}$$

$$[\eta]_d^+ = \begin{cases} \eta, & \text{if } \langle \eta, d \rangle \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

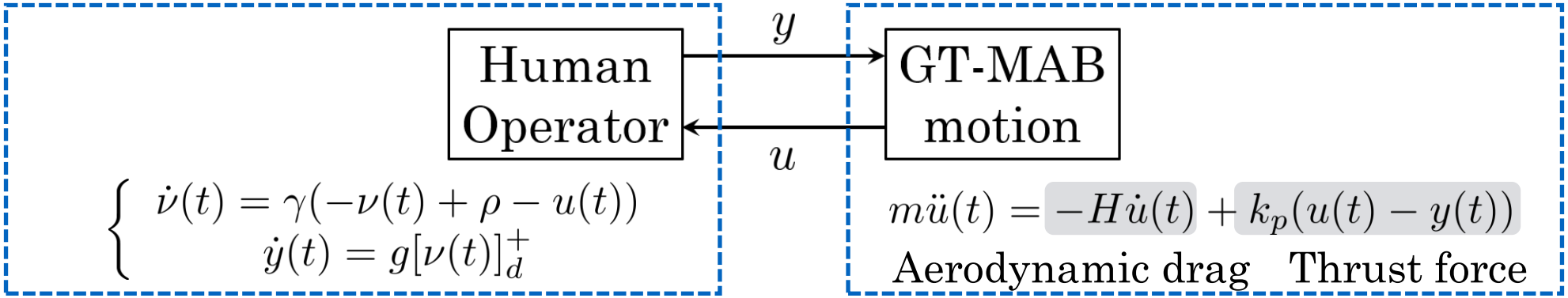
$\eta(t)$: Human internal state

d: Direction from pointer's initial position to target

γ, g: Feedback gain

Bullock D, and Grossberg S. Neural dynamics of planned arm movements: emergent invariants and speed-accuracy properties during trajectory formation. *Psychological Review* 1988; 95(1):49–90.

Stability analysis of the closed-loop system



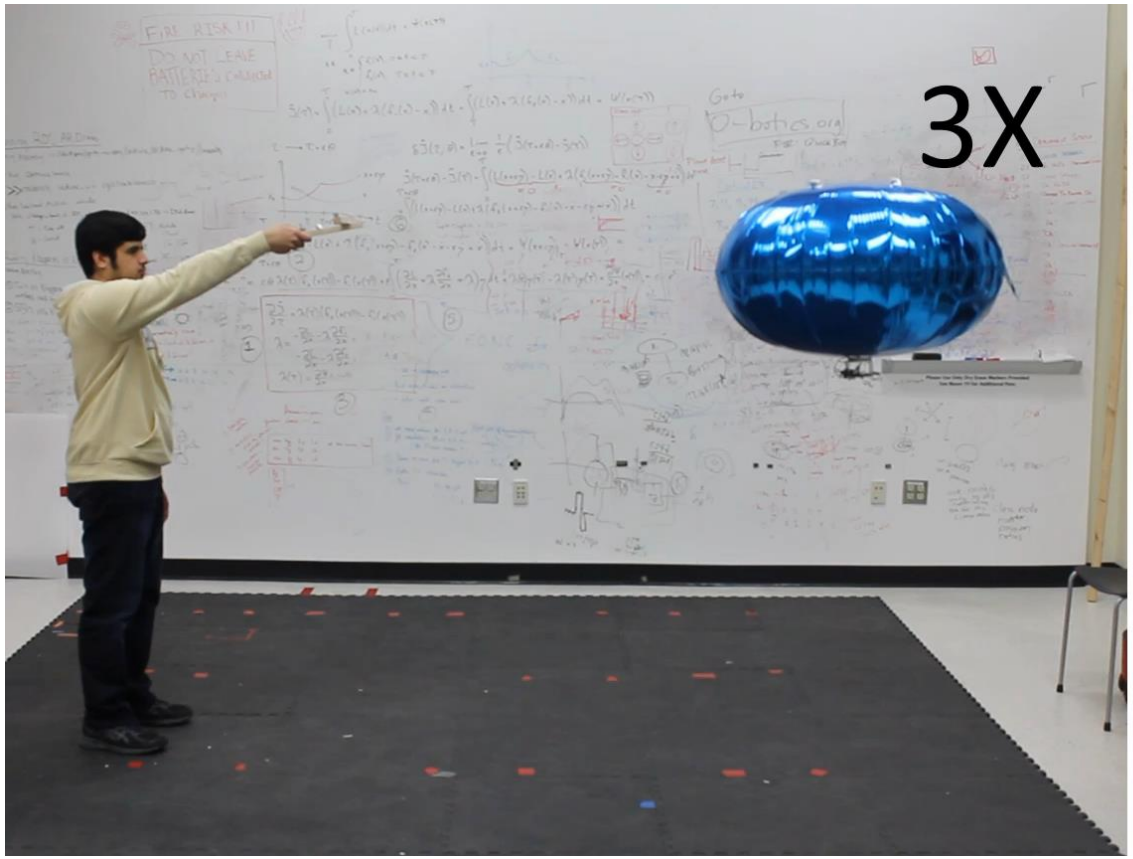
For the stability analysis, we assume

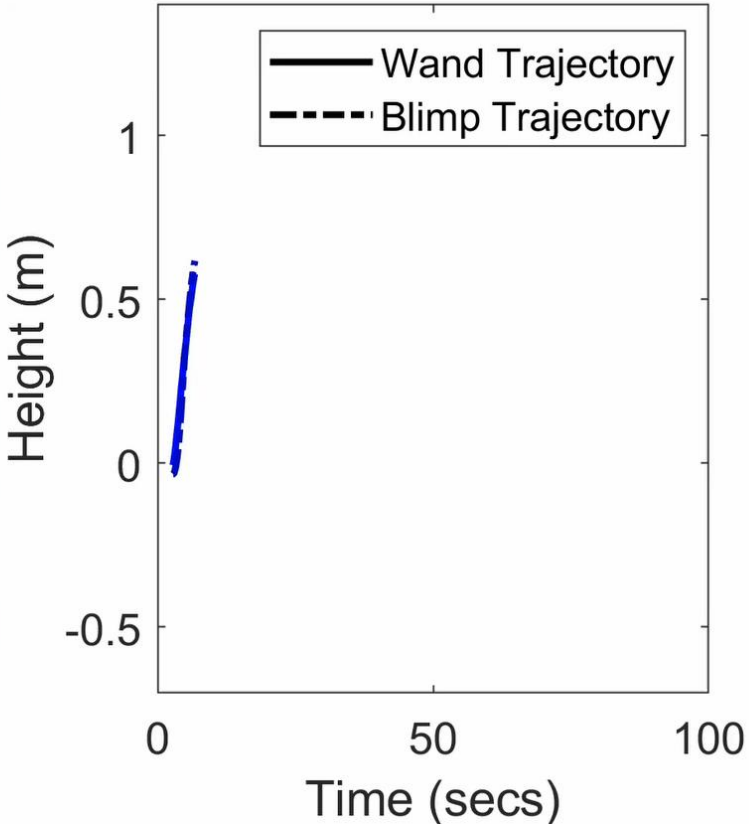
- The target position is at the origin of the coordinate.

Theorem 1. *For the closed-loop system, with* $H, g, \gamma > 0, k_p < \gamma^2(1 - \frac{g}{H}) + \gamma(H - 2g) - gH$, *if the states start from the region* $\eta > 0$, *the states exponentially converge to the equilibrium set*

$$E = \{\dot{u} = 0, \text{ and } u = y = -\eta, \text{ and } \eta \leq 0\}.$$

Experiment results





$$\text{Going up sections: } \Omega_u = \{n \in \mathbb{Z} | \dot{y}(t) > 0, t \in [T_{n-1}, T_n]\}$$

$$\text{Going down sections: } \Omega_d = \{n \in \mathbb{Z} | \dot{y}(t) < 0, t \in [T_{n-1}, T_n]\}$$

Parameter identification

Unknown parameters:

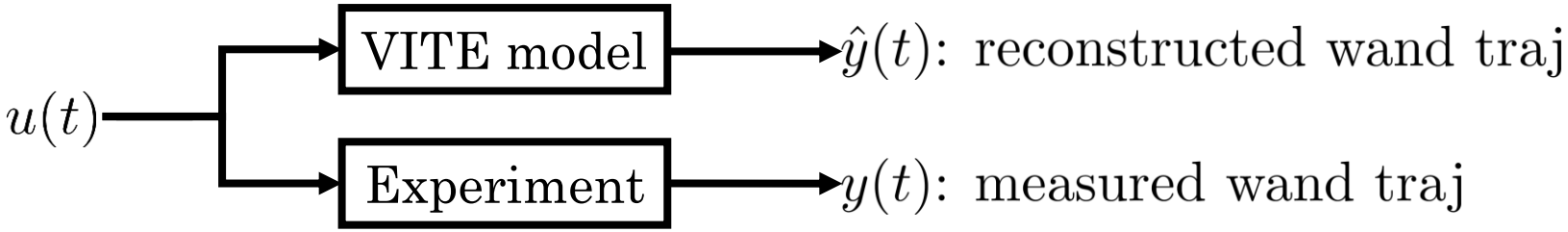
Feedback gain of the VITE model: γ, g

Human intention: ρ

Initial condition of the human internal state: $\eta(T_n)$

VITE model:

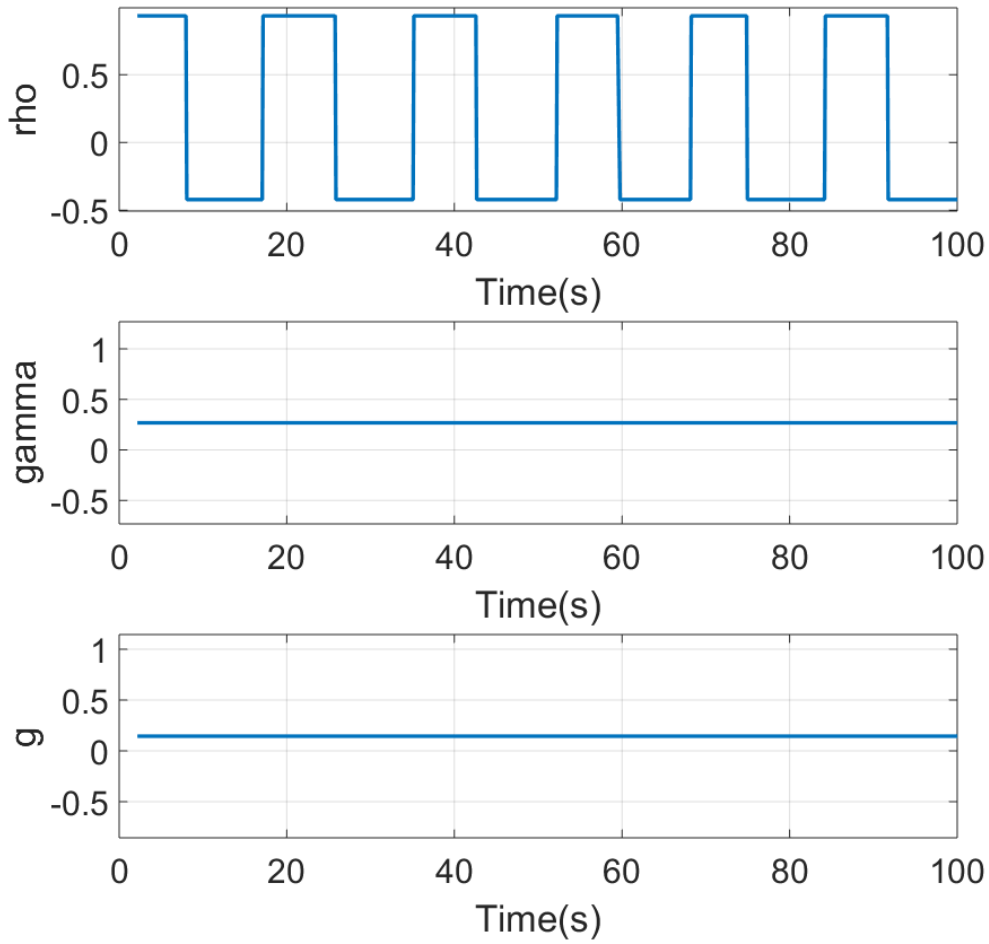
$$\begin{cases} \dot{\eta}(t) = \gamma(-\eta(t) + \rho - u(t)) \\ \dot{y}(t) = g[\eta(t)]_d^+ \end{cases}$$

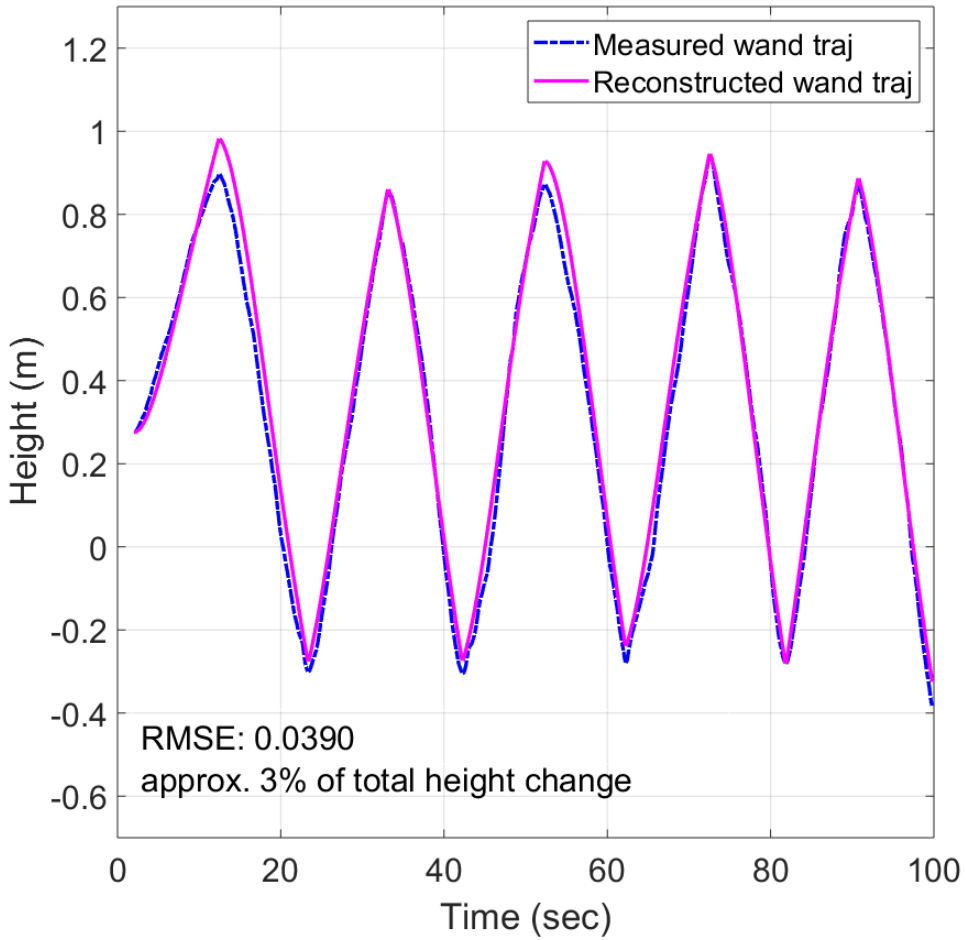


We identify the unknown parameters of the VITE model by solving the following optimization problem:

$$\min \sum_{n \in \Omega_u} \int_{T_{n-1}}^{T_n} (y(t) - \hat{y}(t))^2 dt + \sum_{n \in \Omega_d} \int_{T_{n-1}}^{T_n} (y(t) - \hat{y}(t))^2 dt$$

Experiment results





- The VITE model is applicable to modeling human blimp interaction in 3D space.
- The controller gain satisfies the stability condition in Theorem 1. The closed-loop system is stable

Experiment results

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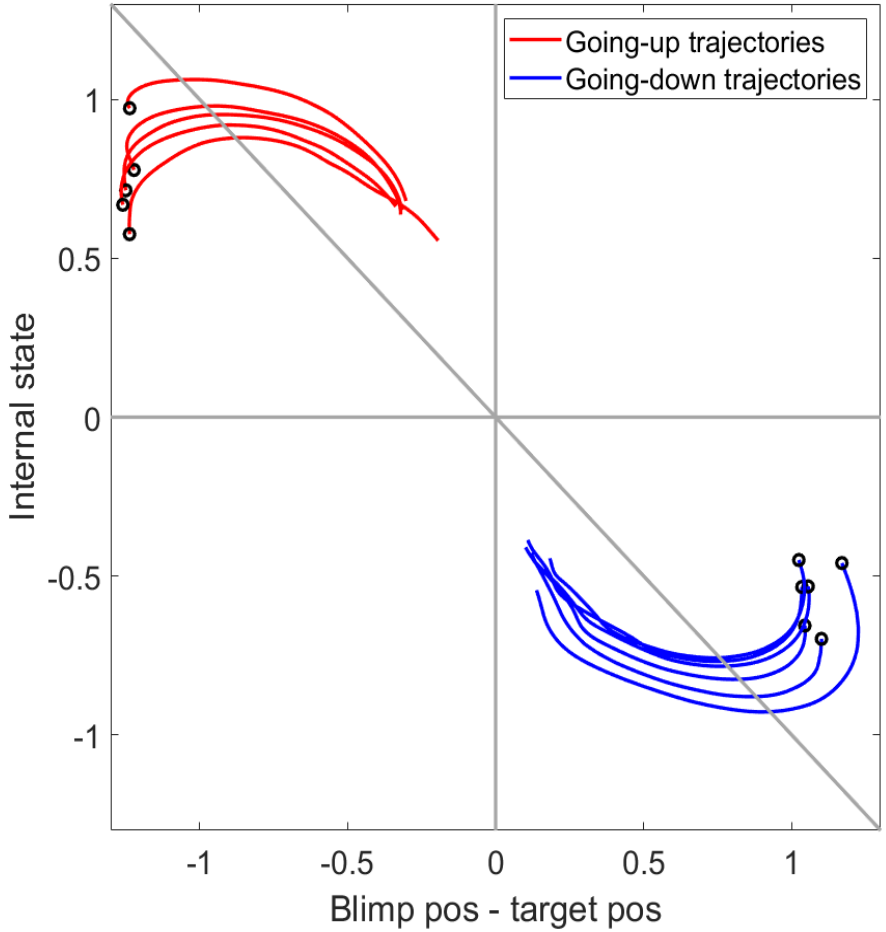
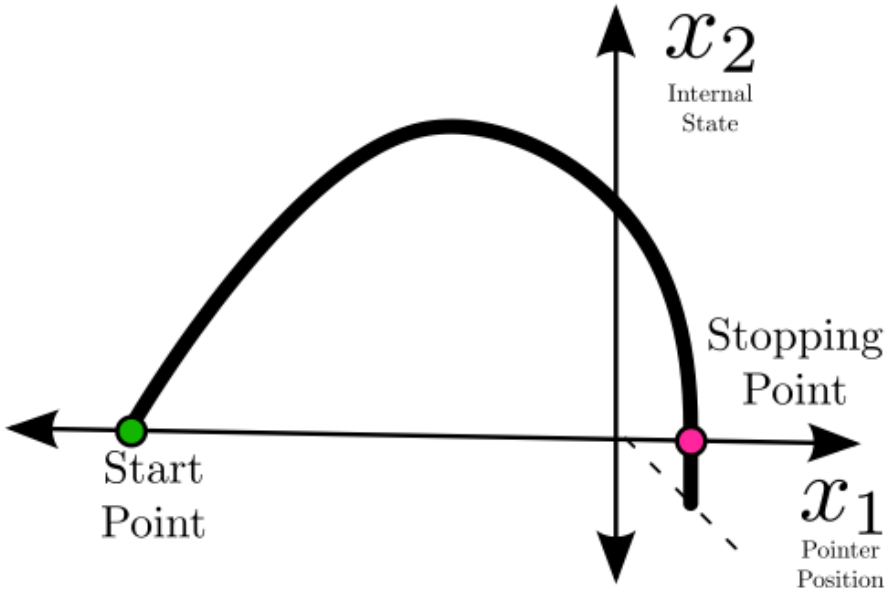


Human-Blimp Interaction: human stops moving the wand before $\eta = 0$

VITE model:

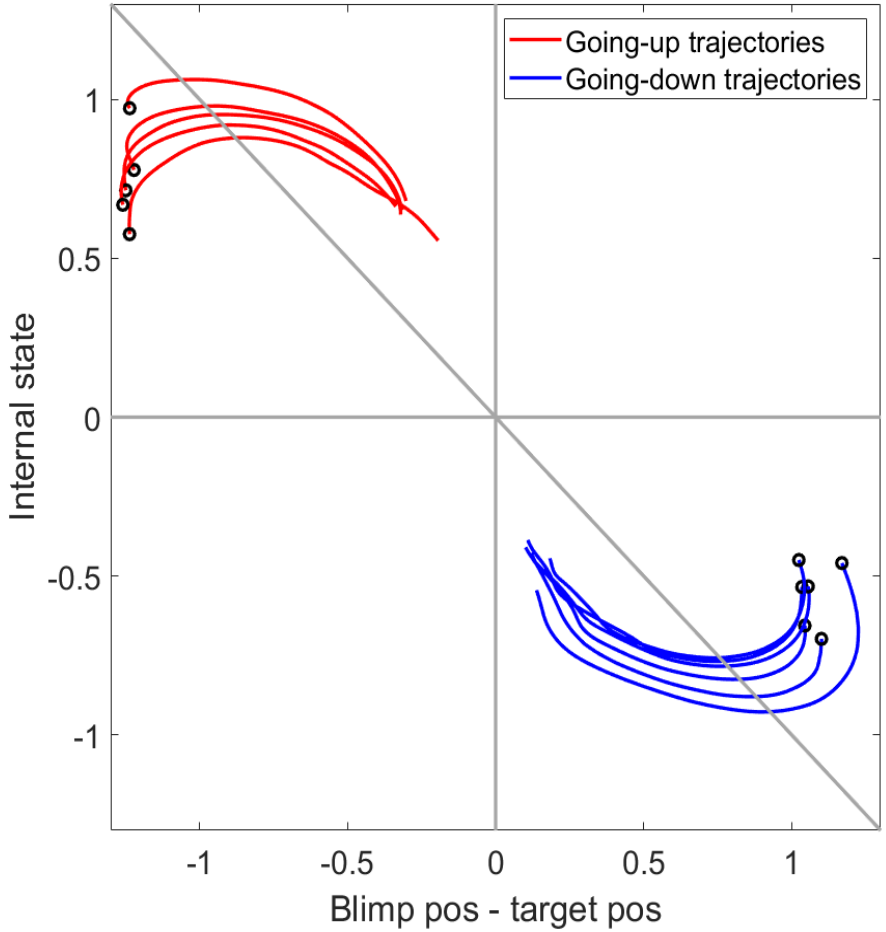
$$\begin{cases} \dot{\eta}(t) = \gamma(-\eta(t) + \rho - u(t)) \\ \dot{y}(t) = g[\eta(t)]_d^+ \end{cases}$$



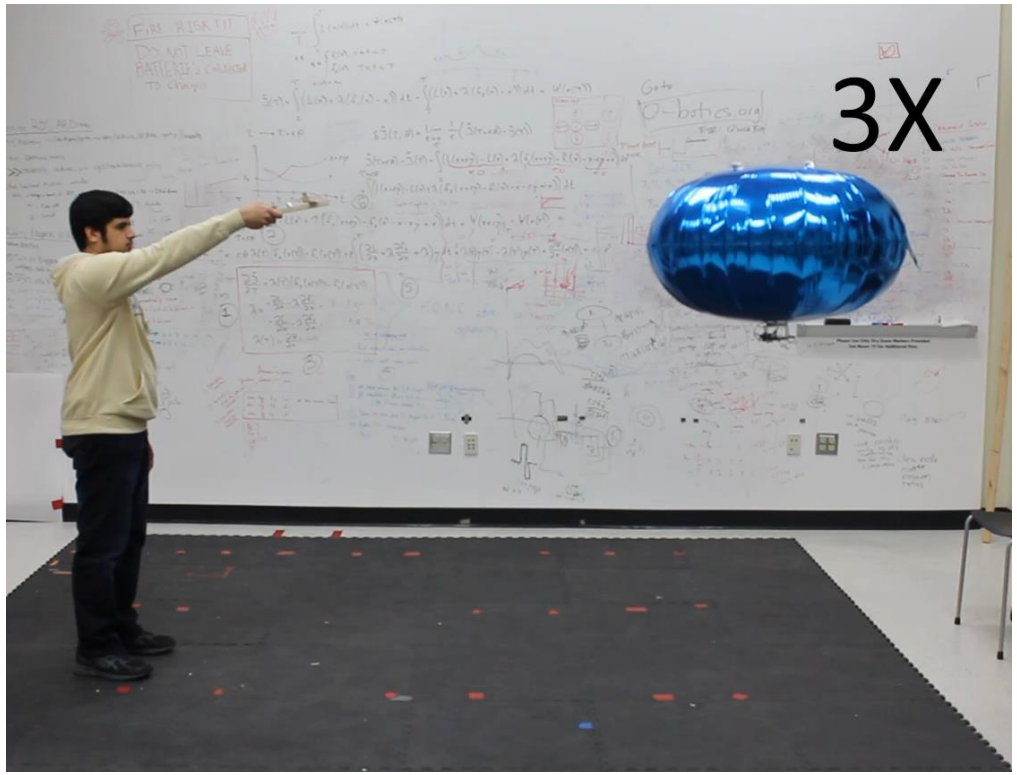
Human-Computer Interaction: human stops moving the pointer when $\eta = 0$

P. Varnell, M. Malisoff, and F. Zhang, "Stability and robustness analysis for human pointing motions with acceleration under feedback delays," International Journal of Robust and Nonlinear Control, vol. 27, no. 5, pp. 703–721, 2016.

Experiment results



Human-Blimp Interaction: human stops moving the wand before $\eta = 0$



Due to GT-MAB's limited thrust force, human stops moving the wand when there is still noticeable difference between blimp & target position.

Conclusion

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- The closed-loop human-blimp interaction dynamics is exponentially stable
- Experimental data verifies that the VITE model is applicable to model human blimp interaction in 3D space
- The human intention can be identified from trajectories of the blimp and pointer movements